Netflix Movies & TV Shows Analysis

1. Project Overview

Introduction

Netflix is one of the world's largest streaming platforms, offering thousands of movies and TV shows across different countries and genres. This project analyzes the Netflix dataset using SQL and Power BI to uncover valuable insights related to content distribution, ratings, genres, release trends, and production countries.

Objectives

- Analyze the distribution of Movies and TV Shows.
- Identify top content-producing countries.
- Explore genre and rating trends.
- Analyze yearly growth of Netflix content.
- Create an interactive Power BI dashboard for visualization.
- Extract business insights using SQL queries.

Tools Used

- SQL
- Power BI
- Excel
- GitHub

Dataset Information

The dataset contains information about Netflix titles, including:

- Title
- Type (Movie/TV Show)
- Director
- Cast
- Country
- Release Year
- Rating
- Duration
- Genre
- Date Added

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2. Data Cleaning & Preparation

Data Cleaning Steps

- Removed duplicate records.
- Handled missing values in Country, Director, and Cast columns.

- Standardized date formats.
- Converted duration values for analysis.
- Verified data consistency across columns.

Database Preparation

The dataset was imported into a SQL database for querying and analysis.

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3. SQL Analysis

SQL Concepts Applied

- SELECT Statements
- Filtering (WHERE)
- Aggregation Functions
- GROUP BY & HAVING
- CASE WHEN
- Subqueries
- Common Table Expressions (CTEs)
- Window Functions

Key Business Questions Solved

1. Total Movies vs TV Shows.
2. Most common content ratings.
3. Top content-producing countries.
4. Most popular genres.
5. Longest movie available on Netflix.
6. Content growth by year.
7. Directors with the highest number of titles.
8. Most frequently appearing actors.
9. Monthly content additions.
10. Country-wise content distribution.

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4. Power BI Dashboard

Dashboard Features

- KPI Cards
- Total Titles
- Total Movies
- Total TV Shows
- Total Countries
- Interactive Filters
- Country
- Rating

- Genre
- Release Year
- Visualizations

- Movies vs TV Shows
- Content by Country
- Genre Distribution
- Release Year Trend
- Rating Analysis
- Monthly Content Additions

Dashboard Benefits

The dashboard enables users to quickly explore Netflix content trends and identify patterns through interactive visualizations.

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5. Key Insights

Content Type Analysis

Movies represent the majority of Netflix content compared to TV Shows.

Country Analysis

A small number of countries contribute a significant portion of Netflix's content library.

Genre Analysis

Drama and International content categories are among the most common genres.

Release Trend Analysis

Netflix content additions increased significantly during recent years, indicating rapid platform expansion.

Rating Analysis

Certain ratings dominate the platform, reflecting Netflix's target audience distribution.

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6. Conclusion

This project demonstrates the use of SQL and Power BI to transform raw data into meaningful business insights. Through data cleaning, SQL analysis, and dashboard development, valuable trends related to Netflix content distribution, ratings, genres, and release patterns were identified.

The project highlights practical skills in:

- Data Cleaning
- SQL Querying
- Data Analysis
- Business Intelligence
- Dashboard Development
- Data Storytelling

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